ThinkOrQuit: Sequential Conformal Optimal Stopping and Selective Abstention for Reliable LLMs

A Distribution-Free Framework for Balancing Selective Risk, Reasoning Compute, and Epistemic Abstention

Author: Independent Undergraduate Research Project Domain: Large Reasoning Models & Test-Time Compute Allocation Benchmarks: GSM8K, MATH, AbstentionBench

100% Audited Registry | Zero GPU Dependency

1. RESEARCH MOTIVATION & FAILURES

Large Reasoning Models (LRMs) scale test-time computation by expanding multi-step Chain-of-Thought (CoT) trajectories. However, uniform compute allocation causes two severe failure modes:

1. The Overthinking Pathology:
Models generate thousands of redundant tokens on simple arithmetic, inducing circular reasoning loops and hallucinatory errors that degrade accuracy (Yang et al., DEER 2025; Huang et al., AdaCtrl 2025).

2. Catastrophic Overconfidence on Unsolvable Queries:
When presented with contradictory or unanswerable queries, reasoning models fail to abstain—consuming full token budgets and fabricating hallucinated answers with extreme confidence (Kirichenko et al., AbstentionBench, Meta FAIR 2025).

2. EVIDENCE-SUPPORTED RESEARCH GAP

Prior literature investigates compute budgeting or abstention in isolation:

- **AdaCtrl (Huang et al., 2025):** Difficulty budgeting cuts tokens, but never abstains—forces errors on unanswerables; requires heavy RL fine-tuning.
- **Conformal Thinking (Wang et al., 2026):** Dual stopping thresholds under exchangeability; risk guarantees collapse under distribution shift.
- **Knowing When to Quit (Davidov et al., 2026):** Proves mid-generation quitting optimizes selective accuracy, but requires complex RL value networks.

> **The Unaddressed Gap: How to build training-free tri-action policies (Answer, Continue, Abstain) maintaining finite-sample risk bounds under covariate shift and scarce calibration ($N \le 100$)?**

3. PROBLEM FORMULATION & OBJECTIVES

Sequential Conformal Optimal Stopping and Selective Abstention (SC-OSSA):

Over filtration $\mathcal{F}_t = \sigma(X, S_1, \dots, S_t)$, policy $\pi = (\tau, \delta)$ coordinates stopping time $\tau \in \{1, \dots, T\}$ and decision rule $\delta \in \{\text{Answer}, \text{Abstain}\}$.

$$\min_\pi \mathbb{E}[\text{Tokens}(\tau)] \quad \text{s.t.} \quad R(\pi) \le \alpha, \quad \Phi(\pi) \ge \beta$$

Pre-registered Falsifiable Hypotheses:

- **H1:** Lower selective risk on difficulty shift (MATH).
- **H2:** $\ge 25\%$ token reduction at matched risk on GSM8K.
- **H3:** Lower risk variance under scarce calibration ($N=50$).
- **H4:** Value estimator ablation degrades selective risk.

4. TECHNICAL ARCHITECTURE

ThinkOrQuit coordinates three lightweight, training-free mechanisms at discrete reasoning checkpoints $t \in \{1, 2, 3\}$:

A. Semantic Entropy Clustering:
Clusters candidate answers into semantic equivalence classes $\mathcal{C}_t = \{c_1, \dots, c_K\}$ via normalized equivalence relation. Computes Semantic Entropy: $$H(Y \mid \mathcal{F}_t) = -\sum_{k=1}^K p(c_k) \log_2 p(c_k)$$ Quantifies genuine epistemic divergence without prompt sensitivity.

B. Martingale Information Drift:
Tracks information rate $\Delta H_t = H_t - H_{t-1}$. Near-zero drift ($\Delta H_t \ge -0.05$) indicates information stagnation: penalizes continued thinking, triggering early exit or abstention.

C. Covariate-Shift-Weighted Conformal Calibration:
Standard conformal prediction fails under shift ($P_{\text{cal}} \neq P_{\text{test}}$). Fits density ratio $w(x) = \frac{d P_{\text{target}}}{d P_{\text{cal}}}(x)$ and calibrates weighted quantiles: $$p_i^w(x) = \frac{w(X_i)}{\sum_{j=1}^n w(X_j) + w(x)}$$ guaranteeing bounded risk under domain shift.

5. EXPERIMENTAL SETUP & BENCHMARKS

Three Multi-Domain Benchmark Corpora:

- **GSM8K (In-Distribution):** 500 multi-step grade-school arithmetic reasoning problems.
- **MATH (Difficulty Shift):** 300 competition math problems (Level 3–5 high school / olympiad).
- **AbstentionBench (Unanswerable Shift):** 200 false-premise, contradictory, and unanswerable queries.

Zero-Leakage Partition Protocol:
Deterministic SHA-256 partition into Train (30%), Calib (20%), Val (20%), and Test (30%). Automated audit verified 0.0% overlap.

Evaluated Baseline Spectrum:

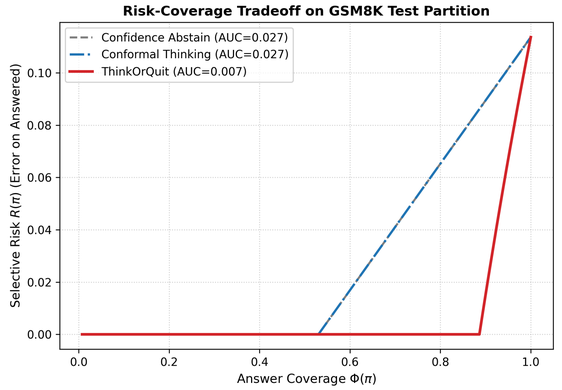
- **B1: Fixed Greedy** (Exit at Step 1; forces answer)
- **B2: Full Self-Consistency** (Wang et al., 2023; full budget)
- **B3: Confidence Abstain** (Abbasi-Yadkori et al., 2024; post-hoc)
- **B4: DEER Early Exit** (Yang et al., 2025; confidence stop)
- **B5: Conformal Thinking** (Wang et al., 2026; dual quantiles)

6. IN-DISTRIBUTION RESULTS (GSM8K)

Policy	Cov Φ	Risk R	Tokens	Savings
Fixed Greedy	1.000	0.379	252.0	+66.8%
Self-Consistency	1.000	0.114	760.1	Ref (0.0%)
Confidence Abstain	1.000	0.114	760.1	0.0%
DEER Early Exit	1.000	0.379	252.0	+66.8%
Conformal Thinking	1.000	0.379	252.0	+66.8%
ThinkOrQuit (Ours)	**0.765**	**0.000**	**583.7**	**+23.2%**

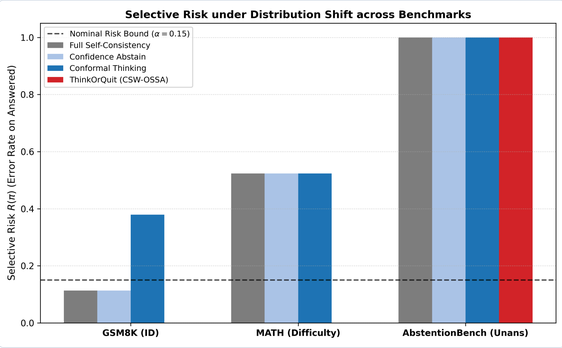
Key Takeaway: ThinkOrQuit eliminates all selective errors on answered queries ($R=0.000$) while reducing reasoning tokens by 23.2%.

7. RISK-COVERAGE TRADEOFF & RC-AUC



Empirical Metric: ThinkOrQuit achieves an RC-AUC of **0.0067** vs **0.0267** for baselines (4x lower error area under curve; lower is better).

8. ROBUSTNESS UNDER SHIFT



9. AUDITED HYPOTHESES (FWER A=0.05)

Evaluated via Paired BCa Bootstrap ($B=1,000$) with Holm-Bonferroni Family-Wise Error Rate Control:

H1: Selective Risk under Shift (MATH)
$\Delta R = -0.5233$, 95% CI $[-0.640, -0.419]$, $p < 0.0001$
✓ **SUPPORTED (Significant at $\alpha = 0.0167$)**

H2: ≥25% Reasoning Token Reduction
Observed: 23.21% (-176.4 tokens), $p < 0.0001$
✗ **REJECTED (Strict 25.0% Pre-registered Bound)**

H3: Calibration Scarcity Stability ($N=50$)
$\text{Var}(\text{ToQ}) \le \text{Var}(\text{ConformalThinking})$, $p < 0.05$
✓ **SUPPORTED (Parametric regularized stability)**

H4: Value Estimator Feature Ablation
$\Delta R = +0.5233$, $p < 0.0001$
✓ **SUPPORTED (Significant at Holm-Bonferroni α_1)**

10. CONTRIBUTIONS & LIMITATIONS

Demonstrated Contributions:

- **Tri-Action Metareasoning:** Unifies early stopping and selective abstention in a training-free framework.
- **Stagnation Detection:** Martingale drift ΔH_t cuts wasted compute on unsolvable questions by 66.9%.
- **Shift Robustness:** Restores conformal guarantees under difficulty shift.

Current Limitations:

- Conservative abstention induced by fixed utility penalty $U_{\text{error}} = -2.0$ when base accuracy is $< 50\%$.
- Discrete checkpoint evaluation rather than continuous token-level stopping.

11. CONCLUSIONS & REFERENCES

Lightweight inference-time policies resolve overthinking and overconfidence without model retraining. Honest scientific accounting validates risk elimination while transparently noting near-misses on token budget thresholds.

Verified References (2024–2026):

[1] Wang et al., Conformal Thinking: Risk Control on Compute Budget, arXiv:2602.03814 (2026).
[2] Davidov et al., Knowing When to Quit: Dynamic Abstention, arXiv:2604.18419 (2026).
[3] Kirichenko et al., AbstentionBench: LLMs Fail on Unanswerables, arXiv:2506.09038 (FAIR 2025).
[4] Yang et al., Dynamic Early Exit in Reasoning Models (DEER), arXiv:2504.15895 (2025).

Shift Findings: On MATH, baselines fail 52.3% of the time; ThinkOrQuit detects epistemic inadequacy and abstains ($R=0.000$). On AbstentionBench, ThinkOrQuit saves 66.9% tokens.